

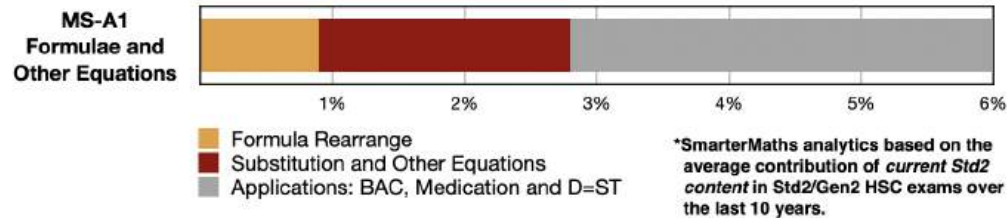
STD 1: Algebra (Std 1), A1 Formulae and Equations (Y11)

Substitution and Other Equations (Std 1)

Teacher: Denis Vlismas

Exam Equivalent Time: 81 minutes (based on HSC allocation of 1.5 minutes approx. per mark)

Gen2/Std2 Exam Contribution History A1 Formulae and Equations



IMPORTANT FEATURES AND TIPS FROM EXAM HISTORY

- MS-A1 *Formulae and Equations* has accounted for 6% and 5% of the 2020 and 2019 Std1 exams respectively.
- MS-A1 *Formulae and Equation* is split into the sub-topics listed in the graph above. Note that estimated exam contributions in the graph are taken from Std2/Gen2 past exams due to the very limited Std1 exam history. They should only be used for estimating the "relative" importance of each sub-topic.
- This analysis looks at the sub-topic *Substitution and Other Equations*.

ANALYSIS - What to Expect and Common pitfalls

- Substitution and Other Equations* were tested twice in the 2020 exam, and once in 2019 (this example required both rearranging and substitution).
- The 2020 Std1 18 exam question that required substitution into a "stopping distance" equation, resulted in a surprisingly low mean mark of 21% and should be reviewed closely.
- Pitfalls:** In Std2/Gen2 exam history, students struggled with algebraic fractions in 2018 Std2 28b and substituting a negative number into an equation 2010 Std2 7 MC. Both warrant revision attention.
- Revision recommended for "Find the Mistake" style questions that have been poorly answered in the past.

Questions

1. Algebra, STD2 A1 2005 HSC 2 MC

What is the value of $\frac{a-b}{4}$, if $a = 240$ and $b = 56$?

- (A) 4
- (B) 46
- (C) 226
- (D) 736

2. Algebra, STD2 A1 2006 HSC 2 MC

If $V = \frac{4}{3}\pi r^3$, what is the value of V when $r = 2$, correct to two decimal places?

- (A) 8.38
- (B) 12.57
- (C) 25.13
- (D) 33.51

3. Algebra, STD2 A1 2016 HSC 2 MC

Which of the following equations has $x = 5$ as the solution?

- (A) $x - 5 = 10$
- (B) $5 - x = 10$
- (C) $\frac{x}{2} = 10$
- (D) $2x = 10$

4. Algebra, STD2 A1 2017 HSC 7 MC

It is given that $I = \frac{3}{2}MR^2$.

What is the value of I when $M = 26.55$ and $R = 3.07$, correct to two decimal places?

- A. 375.35
- B. 3246.08
- C. 9965.45
- D. 14 948.18

5. Algebra, STD2 A1 2004 HSC 3 MC

If $K = Ft^3$, $F = 5$ and $t = 0.715$, what is the value of K correct to three significant figures?

- (A) 1.82
- (B) 1.827
- (C) 1.828
- (D) 1.83

6. Algebra, STD2 A1 2015 HSC 2 MC

Which of the following is $4x + 3y - x - 5y$ in its simplest form?

- (A) $3x - 2y$
- (B) $3x + 8y$
- (C) $5x - 2y$
- (D) $5x + 8y$

7. Algebra, STD2 A1 2017 HSC 9 MC

What is the value of x in the equation $\frac{5-x}{3} = 6$?

- A. -13
- B. -3
- C. 3
- D. 13

8. Algebra, STD2 A1 2008 HSC 9 MC

What is the value of $\sqrt{\frac{x+2y}{8y}}$ if $x = 5.6$ and $y = 3.1$, correct to 2 decimal places?

- (A) 0.69
- (B) 2.62
- (C) 2.83
- (D) 4.77

9. Algebra, STD2 A1 2004 HSC 11 MC

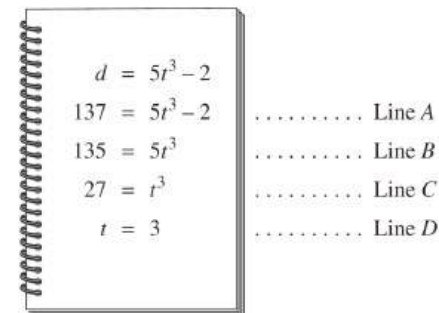
If $d = 6t^2$, what is a possible value of t when $d = 2400$?

- (A) 0.05
- (B) 20
- (C) 120
- (D) 400

10. Algebra, STD2 A1 2005 HSC 14 MC

Using the formula $d = 5t^3 - 2$, Marcia tried to find the value of t when $d = 137$.

Here is her solution. She has made one mistake.



Which line does NOT follow correctly from the previous line?

- (A) Line A
- (B) Line B
- (C) Line C
- (D) Line D

11. Algebra, STD2 A1 2015 HSC 24 MC

Consider the equation $\frac{2x}{3} - 4 = \frac{5x}{2} + 1$.

Which of the following would be a correct step in solving this equation?

- A. $\frac{2x}{3} - 3 = \frac{5x}{2}$
- B. $\frac{2x}{3} = \frac{5x}{2} + 5$
- C. $2x - 4 = \frac{15x}{2} + 3$
- D. $\frac{4x}{6} - 8 = 5x + 2$

12. Algebra, STD2 A1 2016 HSC 5 MC

Which expression is equivalent to $2(3x - 4) + 2$?

- (A) $6x - 2$
- (B) $6x - 4$
- (C) $6x - 6$
- (D) $6x - 10$

13. Algebra, STD1 A1 2021 HSC 6 MC

Blood pressure is recorded as systolic pressure/diastolic pressure and measured in mmHg.

When standing upright, Mary's blood pressure was 139/85. Three minutes after sitting down, her blood pressure was 118/74.

What was the change in Mary's diastolic blood pressure?

- A. There was an increase of 11 mmHg.
- B. There was an increase of 21 mmHg.
- C. There was a decrease of 11 mmHg.
- D. There was a decrease of 21 mmHg.

14. Algebra, STD2 A1 2010 HSC 7 MC

If $M = -9$, what is the value of $\frac{3M^2 + 5M}{6}$?

- (A) -250.5
- (B) -48
- (C) 33
- (D) 235.5

15. Algebra, STD2 A1 SM-Bank 3

Find the value of r given $\frac{r}{7} - 4 = 3$. (1 mark)

16. Algebra, STD2 A1 SM-Bank 13

If $\frac{y-3}{3} = 5$, find y . (1 mark)

17. Algebra, STD2 A1 SM-Bank 2

If $A = P(1+r)^n$, find A given $P = \$300$, $r = 0.12$ and $n = 3$ (give your answer to the nearest cent). (2 marks)

18. Algebra, STD2 A1 2007 HSC 24b

The distance in kilometres (D) of an observer from the centre of a thunderstorm can be estimated by counting the number of seconds (t) between seeing the lightning and first hearing the thunder.

Use the formula $D = \frac{t}{3}$ to estimate the number of seconds between seeing the lightning and hearing the thunder if the storm is 1.2 km away. (1 mark)

19. Algebra, STD2 A1 SM-Bank 1

What is the value of $5a^2 - b$, if $a = -4$ and $b = 3$. (2 marks)

20. Algebra, STD2 A1 SM-Bank 7

If $S = V_0(1-r)^n$, find S given $V_0 = \$42\,000$, $r = 0.16$ and $n = 4$. (give your answer to the nearest cent) (2 marks)

21. Algebra, STD2 A1 SM-Bank 9

The volume of a sphere is given by $V = \frac{4}{3}\pi r^3$ where r is the radius of the sphere.

If the volume of a sphere is 220 cm^3 , find the radius, to 1 decimal place. (3 marks)

22. Algebra, STD2 A1 2009 HSC 25a

Simplify $5 - 2(x + 7)$. (2 marks)

23. Algebra, STD2 A1 2010 HSC 24a

Fred tried to solve this equation and made a mistake in Line 2.

$$\begin{array}{ll} 4(y+2) - 3(y+1) = -3 & \text{Line 1} \\ 4y + 8 - 3y + 3 = -3 & \text{Line 2} \\ y + 11 = -3 & \text{Line 3} \\ y = -14 & \text{Line 4} \end{array}$$

Copy the equation in Line 1 into your writing booklet.

i. Rewrite Line 2 correcting his mistake. (1 mark)

ii. Continue your solution showing the correct working for Lines 3 and 4 to solve this equation for y . (1 mark)

24. Algebra, STD2 A1 2014 HSC 26c

Solve the equation $\frac{5x+1}{3} - 4 = 5 - 7x$. (3 marks)

25. Algebra, STD2 A1 2015 HSC 28d

The formula $C = \frac{5}{9}(F - 32)$ is used to convert temperatures between degrees Fahrenheit (F) and degrees Celsius (C).

Convert 3°C to the equivalent temperature in Fahrenheit. (2 marks)

26. Algebra, STD2 A1 SM-Bank 8

What is the value of $\frac{a+b}{ab}$ if $a = -2.1$ and $b = -3.6$, correct to 1 decimal place? (2 marks)

27. Algebra, STD1 A1 2020 HSC 16

Consider the equation $m = 6 - \frac{3R}{2R-5}$.

Find the value of m when $R = 10$. (2 marks)

28. Algebra, STD2 A1 SM-Bank 10

For adults (18 years and older), the Body Mass Index is given by

$B = \frac{m}{h^2}$ where $m = \text{mass}$ in kilograms and $h = \text{height}$ in metres.

The medically accepted healthy range for B is $21 \leq B \leq 25$.

What is the minimum weight for a 163 cm adult female to be considered healthy? (2 marks)

29. Algebra, STD2 A1 2013 HSC 29a

Sarah tried to solve this equation and made a mistake in Line 2.

$$\begin{array}{ll} \frac{W+4}{3} - \frac{2W-1}{5} = 1 & \text{Line 1} \\ 5W + 20 - 6W - 3 = 15 & \text{Line 2} \\ 17 - W = 15 & \text{Line 3} \\ W = 2 & \text{Line 4} \end{array}$$

Copy the equation in Line 1 into your writing booklet and continue your solution to solve this equation for W .

Show all lines of working. (2 marks)

30. Algebra, STD2 A1 2018 HSC 28b

Solve the equation $\frac{2x}{5} + 1 = \frac{3x+1}{2}$, leaving your answer as a fraction. (3 marks)

31. Algebra, STD1 A1 2019 HSC 34

Given the formula $C = \frac{A(y+1)}{24}$, calculate the value of y when $C = 120$ and $A = 500$. (3 marks)

32. Algebra, STD1 A1 2020 HSC 18

The distance, d metres, travelled by a car slowing down from u km/h to v km/h can be obtained using the formula

$$v^2 = u^2 - 100d$$

What distance does a car travel while slowing down from 70 km/h to 40 km/h? (2 marks)

33. Algebra, STD2 A1 2021 HSC 29

Solve $x + \frac{x-1}{2} = 9$ (2 marks)

Worked Solutions

1. Algebra, STD2 A1 2005 HSC 2 MC

$$\begin{aligned}\frac{a-b}{4} &= \frac{240-56}{4} \\ &= 46 \\ \Rightarrow B\end{aligned}$$

2. Algebra, STD2 A1 2006 HSC 2 MC

$$\begin{aligned}V &= \frac{4}{3}\pi r^3 \\ \text{When } r &= 2, \\ V &= \frac{4}{3}\pi \times 2^3 \\ &= 33.510\dots \\ \Rightarrow D\end{aligned}$$

3. Algebra, STD2 A1 2016 HSC 2 MC

$$\begin{aligned}2x &= 10 \\ \therefore x &= 5 \\ \Rightarrow D\end{aligned}$$

4. Algebra, STD2 A1 2017 HSC 7 MC

$$\begin{aligned}I &= \frac{3}{2} \times 26.55 \times (3.07)^2 \\ &= 375.346\dots \\ \Rightarrow A\end{aligned}$$

5. Algebra, STD2 A1 2004 HSC 3 MC

$$\begin{aligned}K &= Ft^3 \\&= 5 \times (0.715)^3 \\&= 1.8276\dots \\&= 1.83 \text{ (3 sig figures)}\end{aligned}$$

$\Rightarrow D$

6. Algebra, STD2 A1 2015 HSC 2 MC

$$\begin{aligned}4x + 3y - x - 5y \\&= 3x - 2y\end{aligned}$$

$\Rightarrow A$

7. Algebra, STD2 A1 2017 HSC 9 MC

$$\begin{aligned}\frac{5-x}{3} &= 6 \\5-x &= 18 \\x &= 5-18 \\&= -13\end{aligned}$$

$\Rightarrow A$

8. Algebra, STD2 A1 2008 HSC 9 MC

$$\begin{aligned}\sqrt{\frac{x+2y}{8y}} &= \sqrt{\frac{5.6 + (2 \times 3.1)}{(8 \times 3.1)}} \\&= \sqrt{\frac{11.8}{24.8}} \\&= 0.6897\dots\end{aligned}$$

$\Rightarrow A$

9. Algebra, STD2 A1 2004 HSC 11 MC

$$\begin{aligned}d &= 6t^2 \\t^2 &= \frac{d}{6} \\t &= \pm \sqrt{\frac{d}{6}}\end{aligned}$$

When $d = 2400$

$$\begin{aligned}t &= \pm \sqrt{\frac{2400}{6}} \\&= \pm 20\end{aligned}$$

$\Rightarrow B$

10. Algebra, STD2 A1 2005 HSC 14 MC

$$\begin{aligned}d &= 5t^3 - 2 \\137 &= 5t^3 - 2 \dots \text{Line A} \\139 &= 5t^3 \dots \text{Line B}\end{aligned}$$

$\therefore$ Line B doesn't follow on correctly.

$\Rightarrow B$

11. Algebra, STD2 A1 2015 HSC 24 MC

$$\begin{aligned}\frac{2x}{3} - 4 &= \frac{5x}{2} + 1 \\ \frac{2x}{3} &= \frac{5x}{2} + 5\end{aligned}$$

$\Rightarrow B$

12. Algebra, STD2 A1 2016 HSC 5 MC

$$\begin{aligned}2(3x - 4) + 2 \\&= 6x - 8 + 2 \\&= 6x - 6\end{aligned}$$

$\Rightarrow C$

13. Algebra, STD1 A1 2021 HSC 6 MC

Diastolic pressure went from 85 to 74,
or decreased by 11 mmHg.

$\Rightarrow C$

♦ Mean mark 45%.

14. Algebra, STD2 A1 2010 HSC 7 MC

$$\begin{aligned}\frac{3M^2 + 5M}{6} &= \frac{3 \times (-9)^2 + 5 \times (-9)}{6} \\ &= \frac{(3 \times 81) - 45}{6} \\ &= \frac{198}{6} \\ &= 33\end{aligned}$$

$\Rightarrow C$

♦♦ Only 31% of students answered correctly!

15. Algebra, STD2 A1 SM-Bank 3

$$\begin{aligned}\frac{r}{7} - 4 &= 3 \\ \frac{r}{7} &= 7 \\ \therefore r &= 49\end{aligned}$$

16. Algebra, STD2 A1 SM-Bank 13

$$\begin{aligned}\frac{y-3}{3} &= 5 \\ y-3 &= 15 \\ y &= 18\end{aligned}$$

17. Algebra, STD2 A1 SM-Bank 2

$$\begin{aligned}A &= P(1+r)^n \\ &= 300(1+0.12)^3 \\ &= 300(1.12)^3 \\ &= 421.478... \\ &= \$421.48 \text{ (nearest cent)}\end{aligned}$$

18. Algebra, STD2 A1 2007 HSC 24b

$$D = \frac{t}{3}$$

When $D = 1.2$,

$$\frac{t}{3} = 1.2$$

$$t = 3.6 \text{ seconds}$$

19. Algebra, STD2 A1 SM-Bank 1

$$\begin{aligned}5a^2 - b &= 5(-4)^2 - 3 \\ &= 5 \times 16 - 3 \\ &= 77\end{aligned}$$

20. Algebra, STD2 A1 SM-Bank 7

$$\begin{aligned}S &= V_0(1-r)^n \\ &= 42\,000(1-0.16)^4 \\ &= 42\,000(0.84)^4 \\ &= \$20\,910.597... \\ &= \$20\,910.60 \text{ (to nearest cent)}\end{aligned}$$

21. Algebra, STD2 A1 SM-Bank 9

$$V = \frac{4}{3}\pi r^3$$

$$3V = 4\pi r^3$$

$$r^3 = \frac{3V}{4\pi}$$

When $V = 220$

$$r^3 = \frac{3 \times 220}{4\pi}$$

$$= 52.521\dots$$

$$\therefore r = \sqrt[3]{52.521\dots}$$

$$= 3.744\dots \text{ (by calc)}$$

$$= 3.7 \text{ cm (to 1 d.p.)}$$

22. Algebra, STD2 A1 2009 HSC 25a

$$\begin{aligned} 5 - 2(x + 7) &= 5 - 2x - 14 \\ &= -2x - 9 \end{aligned}$$

♦ Mean mark 47%

23. Algebra, STD2 A1 2010 HSC 24a

i. $4(y + 2) - 3(y + 1) = -3$ **Line 1**

$$4y + 8 - 3y - 3 = -3 \quad \text{Line 2}$$

ii. $y + 5 = -3$ **Line 3**

$$y = -8 \quad \text{Line 4}$$

24. Algebra, STD2 A1 2014 HSC 26c

$$\frac{5x + 1}{3} - 4 = 5 - 7x$$

$$5x + 1 - 3(4) = 3(5 - 7x)$$

$$5x + 1 - 12 = 15 - 21x$$

$$26x = 26$$

$$\therefore x = 1$$

25. Algebra, STD2 A1 2015 HSC 28d

$$C = \frac{5}{9}(F - 32)$$

$$F - 32 = \frac{9}{5}C$$

$$F = \frac{9}{5}C + 32$$

When $C = 3$,

$$F = \left(\frac{9}{5} \times 3\right) + 32$$

$$= 37.4 \text{ degrees } F$$

26. Algebra, STD2 A1 SM-Bank 8

$$\frac{a + b}{ab} = \frac{-2.1 - 3.6}{-2.1 \times -3.6}$$

$$= \frac{-5.7}{7.56}$$

$$= -0.753\dots$$

$$= -0.8$$

27. Algebra, STD1 A1 2020 HSC 16

$$\begin{aligned} m &= 6 - \frac{3 \times 10}{2 \times 10 - 5} \\ &= 6 - \frac{30}{15} \\ &= 6 - 2 \\ &= 4 \end{aligned}$$

28. Algebra, STD2 A1 SM-Bank 10

$$\begin{aligned} B &= \frac{m}{h^2} \\ h &= 163 \text{ cm} = 1.63 \text{ m} \end{aligned}$$

Given $21 \leq B \leq 25$,
 $\Rightarrow B = 21$ for minimum healthy weight.

$$\begin{aligned} 21 &= \frac{m}{1.63^2} \\ \therefore m &= 21 \times 1.63^2 \\ &= 55.794... \\ &= 55.8 \text{ kg (1 d.p.)} \end{aligned}$$

29. Algebra, STD2 A1 2013 HSC 29a

$$\begin{aligned} \frac{W+4}{3} - \frac{2W-1}{5} &= 1 && \dots \text{Line 1} \\ 5W+20-6W+3 &= 15 && \dots \text{Line 2} \\ 23-W &= 15 && \dots \text{Line 2} \\ W &= 8 && \dots \text{Line 4} \end{aligned}$$

♦♦ Mean mark 27%
STRATEGY: The RHS of the equation increases from 1 to 15 (from Line 1 to Line 2), indicating both sides must have been multiplied by 15.

30. Algebra, STD2 A1 2018 HSC 28b

$$\begin{aligned} \underbrace{\frac{2x}{5} + 1}_{\text{multiply } \times 10} &= \underbrace{\frac{3x+1}{2}}_{\text{multiply } \times 10} \\ 4x + 10 &= 15x + 5 \\ 11x &= 5 \\ x &= \frac{5}{11} \end{aligned}$$

♦ Mean mark 35%.

31. Algebra, STD1 A1 2019 HSC 34

Make y the subject:

$$\begin{aligned} C &= \frac{A(y+1)}{24} \\ 24C &= A(y+1) \\ y+1 &= \frac{24C}{A} \\ y &= \frac{24C}{A} - 1 \\ &= \frac{24 \times 120}{500} - 1 \\ &= 4.76 \end{aligned}$$

♦♦ Mean mark 32% (Std1).

32. Algebra, STD1 A1 2020 HSC 18

$$\begin{aligned} u &= 70, v = 40 \\ v^2 &= u^2 - 100d \\ 40^2 &= 70^2 - 100d \\ 100d &= 70^2 - 40^2 \\ \therefore d &= \frac{70^2 - 40^2}{100} \\ &= 33 \text{ metres} \end{aligned}$$

♦♦ Mean mark 21%.

$$x + \frac{x-1}{2} = 9$$

$$2x + x - 1 = 18$$

$$3x = 19$$

$$x = \frac{19}{3}$$

♦ Mean mark 40%.